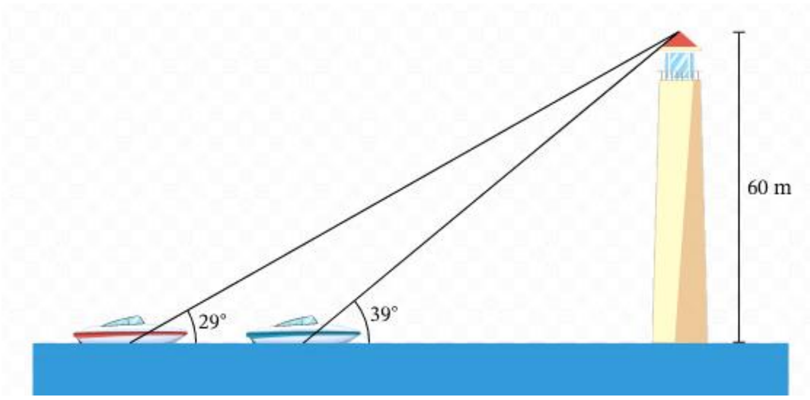
1. There are TWO boats behind each other at sea. The angle of elevation from the **first** boat to the top of a lighthouse, 60 metres tall, is 29°. In addition, the angle of elevation from the **second** boat is 39°, as shown in the diagram.



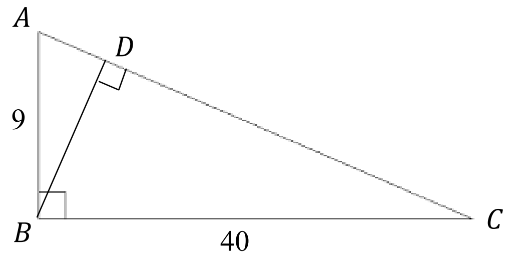
(*a*) Calculate the distance between the base of the lighthouse and the **first** boat, correct to 1 decimal place. **2**

(*b*) Similarly, calculate the distance between the base of the lighthouse and the **second** boat, correct to 1 decimal place. **1**

(*c*) Hence, calculate the distance between the two boats, correct to 1 decimal place. **1**

Section 1 continues on the following page.

2. ABC is a right-angled triangle where $\angle ABC = 90°$ and $BD \perp AC$. The lengths of AB and BC are 9 and 40 respectively.



(*a*) Calculate $\angle C$ to the nearest minute. **1**

(*b*) Hence, or otherwise, calculate the length of BD, correct to 1 decimal place. **2**

3. Let the bearing of B from A be a reflex angle θ.

(Hint: A reflex angle is the angle between 180° and 360°.)

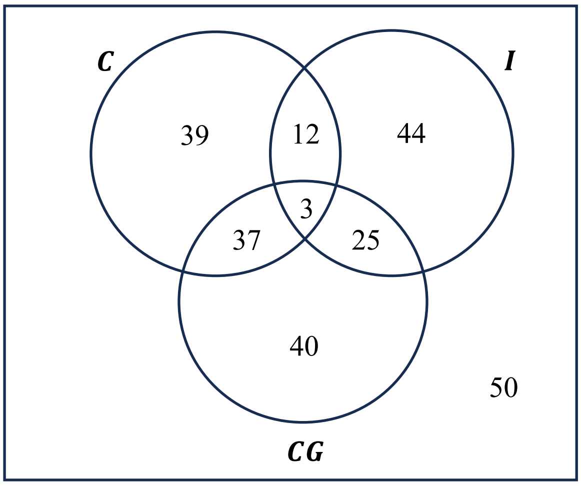
(*a*) By drawing a diagram, find the bearing of A from B in terms of θ. **2**

(*d*) Hence, find the difference between the two bearings. **1**

End of Section 1

1. A high school offers Chinese ($\boldsymbol{C}$), Classical Greek ($\boldsymbol{CG}$) and Italian ($\boldsymbol{I}$) as language subjects for year 12 students for their HSC.

The last year's subject selection was shown in the following Venn diagram.



(***a***) How many year 12 students were studying at the high school last year? **1**

(***b***) Calculate the probability of selecting a student who studied at least one language subject. **1**

(***c***) Calculate the probability of selecting a student who studied Chinese **or** Classical Greek. **1**

(***d***) Calculate the probability of selecting a student who studied Classical Greek **and** Italian. **1**

(***e***) Given that a student studied at least two language subjects, calculate the probability that the student studied Chinese. **1**

Section 2 continues on the following page.

2. From a set of cards numbered 0 to 4, three cards are drawn at random to form a three-digit **odd** number.

(Hint: Examples of odd numbers are 1, 3, 5, 7, 9, 11, 13, . . .)

(***a***) Show that there are 18 possible outcomes for the above event, using a tree diagram. **2**

(***b***) Calculate the probability of having a prime number. **1**

(Hint: A prime number is the number that is divisible by 1 and itself only.)

(***c***) Calculate the probability of having a number whose digits are in descending order. **1**

(***d***) Calculate the probability of having a **prime** number whose digits are in descending order. **1**

End of Section 2

1. A movie received feedback scores from the audience using the scale of 1 (not interesting at all) to 5 (very interesting). The scores were as follows.

2	3	4	5	4
4	5	3	2	1
3	4	2	4	5

(***a***) Fill in **all** the blanks in the following frequency table. **2**

score (x)	frequency (f)	fx
1		
2		
3		
4		
5		
total		

(***b***) Using the table in (***a***), find the mode and the mean of the scores. **2**

(***c***) Which of the following options best describes the types of the above data? **1**

① categorical, discrete ② categorical, continuous

③ numerical, discrete ④ numerical, continuous

Section 3 continues on the following page.

2. Two examination (half-yearly and yearly) marks of 20 randomly chosen students were compared using a back-to-back stem-and-leaf plot.

Half-yearly		Yearly
9 6 3 2 0	5	5 8
8 6 5 3 4 2	6	3 4 9
7 5 4 2	7	0 2 3 6 8
8 3 0	8	2 4 5 7 7
1 0	9	1 5 9
	10	0 0

(***a***) Calculate the mean of the **half-yearly** examination marks. **1**

(***b***) Calculate the median of the **yearly** examination marks. **1**

(***c***) Describe the shape of each examination marks. **2**

(***d***) There is a mistake in the above plot. Circle it on the above plot and briefly explain why. **1**

End of Section 3

1. **(*a*)** Solve the following simultaneous equations, using the **elimination** method. **3**

$$3x - 4y = 41$$
$$2x + 3y = 152$$

(*b*) Solve the following simultaneous equations, using the **substitution** method. **3**

$$2y - 3x = 8$$
$$x + 3y = 1$$

Section 4 continues on the following page.

2. Brian is currently 9 times as old as Alfred. **3**

In 4 years, Brian will be 5 times as old as Alfred.

Calculate their current ages.

(Hint: Let A be the current Alfred's age and B be the current Brian's age.)

Section 4 continues on the following page.

3. The tickets for a music concert are $\$x$ for an adult and $\$y$ for a child. **3**

The tickets of 3 adults and 17 children costs \$123 less than those of 12 adults and 5 children.

If the tickets of 6 adults and 9 children costs \$456, how much will it cost to buy the tickets of **23 adults** and **47 children** for the concert?

End of Section 4